

What is Covariance?

Covariance is a statistical relationship between two random variables, showing how they change relative to each other with time. Accordingly, covariance can be of two types: positive and negative.

Types of Covariance:

Positive Covariance –

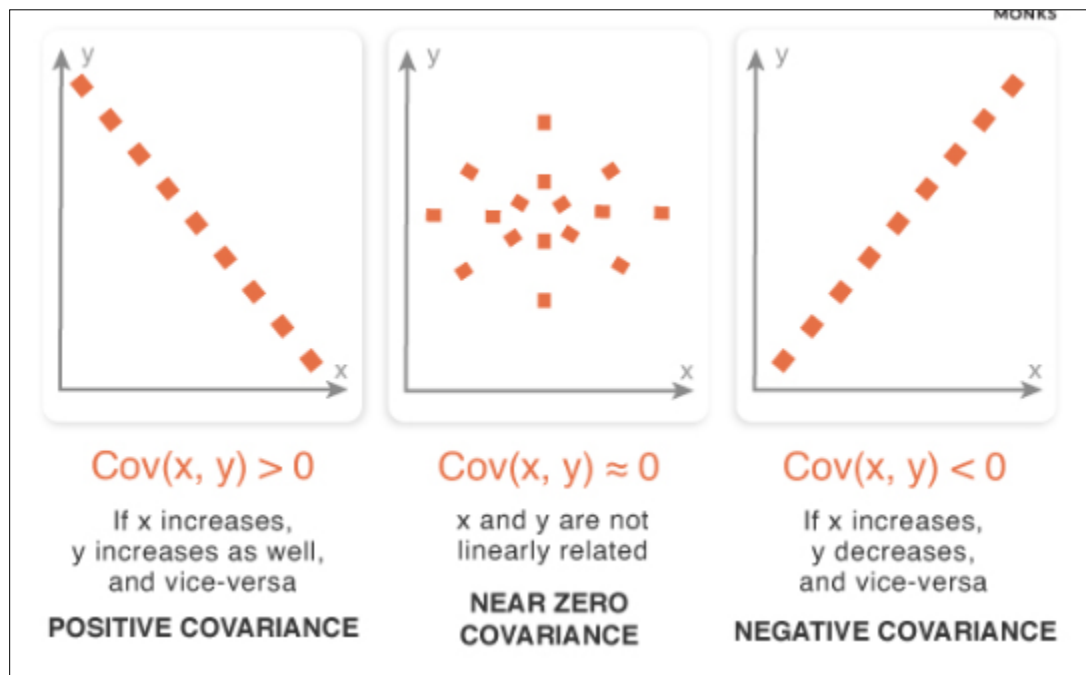
When two variables have positive covariance, they move in the same direction. This means if one variable increases, the other variable also increases, and vice versa.

Negative Covariance-

When two variables have negative covariance, they move in opposite directions. This means if one variable increases, the other tends to decrease, and vice versa.

Zero Covariance:

A covariance close to zero means the variables are not linearly related.



Assignment as per the dataset:

What is the correlation between Degree pass mark and Entrance test pass mark:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
sl_no	3870.000000	-52.641355	-59.598879	-41.465047	52.556168	8.102336	1.138318e+04
ssc_p	-52.641355	117.228377	58.853253	42.702550	37.659225	24.535952	9.088585e+05
hsc_p	-59.598879	58.853253	112.063731	33.684453	33.838355	21.517688	7.310079e+05
degree_p	-41.465047	42.702550	33.684453	53.604710	22.078774	17.185200	4.663363e+05
etest_p	52.556168	37.659225	33.838355	22.078774	176.251018	16.886973	3.727004e+05
mba_p	8.102336	24.535952	21.517688	17.185200	16.886973	34.028376	1.239934e+05
salary	11383.177570	908858.485818	731007.850848	466336.264888	372700.449468	123993.387361	2.259185e+10

- As per the dataset it shows the correlation of degree pass mark and Entrance test pass mark is **22.078774** which indicate the **positive covariance value**. So when degree pass mark increase then etest mark also will increase.

What is Correlation?

Correlation is a statistical measure expressing the strength and direction of the relationship between two or more variables, indicating how they change together.

Types:

Positive Correlation-

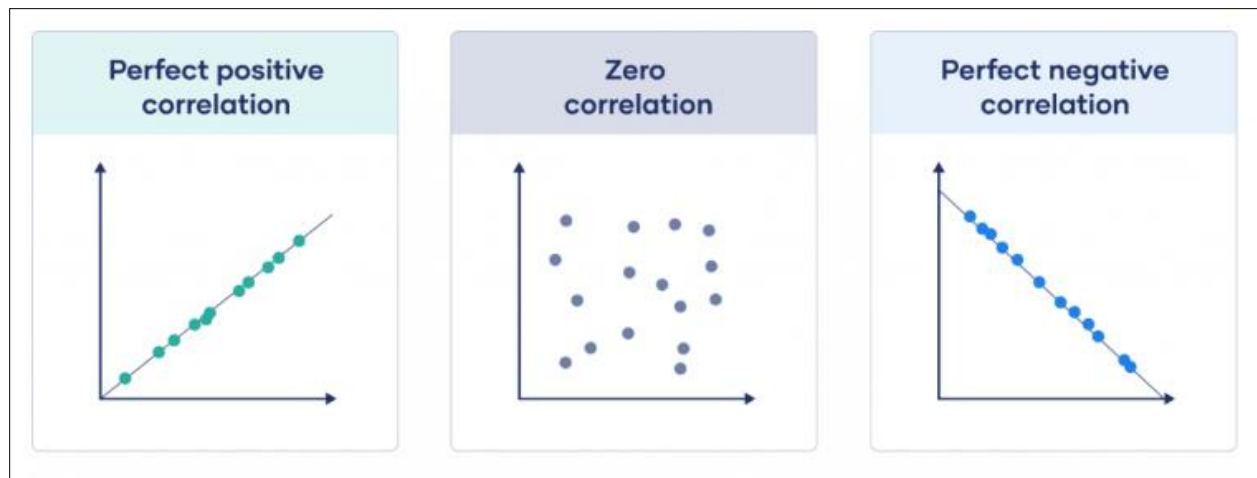
Both variables increase or decrease together

Negative Correlation-

One variable increases while the other decreases

No Correlation-

No apparent relationship exists between the variables.



	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
sl_no	1.000000	-0.078155	-0.090500	-0.091039	0.063636	0.022327	0.001217
ssc_p	-0.078155	1.000000	0.513478	0.538686	0.261993	0.388478	0.558475
hsc_p	-0.090500	0.513478	1.000000	0.434606	0.240775	0.348452	0.459424
degree_p	-0.091039	0.538686	0.434606	1.000000	0.227147	0.402376	0.423762
etest_p	0.063636	0.261993	0.240775	0.227147	1.000000	0.218055	0.186775
mba_p	0.022327	0.388478	0.348452	0.402376	0.218055	1.000000	0.141417
salary	0.001217	0.558475	0.459424	0.423762	0.186775	0.141417	1.000000

Difference between etest_p and mba_p is 0.218 which means when etest mark increases then automatically mba_p also will get increased.

Whereas Degree pass mark and Entrance pass mark difference is 0.22, if degree mark increase then entrance mark also will get increase.